

SET THEORY AND ALGEBRA

(1 Marks)

1. GATE CSE 2026 Set 2

MCQ (More than One Correct Answer) Marks: +1 / -0

Let R be a binary relation on the set $\{1, 2, \dots, 10\}$, where $(x, y) \in R$ if the product of x and y is square of an integer. Which of the following properties is/are satisfied by R ?

- (A) Reflexive
- (B) Symmetric
- (C) Transitive
- (D) Antisymmetric

2. GATE CSE 2025 Set 1

MCQ (Single Correct Answer) Marks: +1 / -0.33

$g(\cdot)$ is a function from A to B , $f(\cdot)$ is a function from B to C , and their composition defined as $f(g(\cdot))$ is a mapping from A to C .

If $f(\cdot)$ and $f(g(\cdot))$ are onto (surjective) functions, which ONE of the following is TRUE about the function $g(\cdot)$?

- (A) $g(\cdot)$ must be an onto (surjective) function.
- (B) $g(\cdot)$ must be a one-to-one (injective) function.
- (C) $g(\cdot)$ must be a bijective function, that is, both one-to-one and onto.
- (D) $g(\cdot)$ is not required to be a one-to-one or onto function.

3. GATE CSE 2024 Set 2

Numerical Marks: +1 / -0

Let P be the partial order defined on the set $\{1, 2, 3, 4\}$ as follows:

$$P = \{(x, x) \mid x \in \{1, 2, 3, 4\}\} \cup \{(1, 2), (3, 2), (3, 4)\}$$

The number of total orders on $\{1, 2, 3, 4\}$ that contain P is _____.

4. GATE CSE 2024 Set 1

Numerical Marks: +1 / -0

Let A and B be non-empty finite sets such that there exist one-to-one and onto functions (i) from A to B and (ii) from $A \times A$ to $A \cup B$. The number of possible values of $|A|$ is _____.

5. GATE CSE 2021 Set 2

MCQ (More than One Correct Answer) Marks: +1 / -0

Consider the following sets, where $n > 2$:

S_1 : Set of all $n \times n$ matrices with entries from the set $\{a, b, c\}$

S_2 : Set of all functions from the set $\{0, 1, 2, \dots, n^2 - 1\}$ to the set $\{0, 1, 2\}$

Which of the following choice(s) is/are correct?

- (A) There does not exist an injection from S_1 to S_2 .
- (B) There exists a bijection from S_1 to S_2 .
- (C) There exists a surjection from S_1 to S_2 .
- (D) There does not exist a bijection from S_1 to S_2 .

6. GATE CSE 2020

Numerical Marks: +1 / -0

Let G be a group of 35 elements. Then the largest possible size of a subgroup of G other than G itself is _____.

7. GATE CSE 2020

Numerical Marks: +1 / -0

Let R be the set of all binary relations on the set $\{1, 2, 3\}$. Suppose a relation is chosen from R at random. The probability that the chosen relation is reflexive (round off to 3 decimal places) is _____.

8. GATE CSE 2019

MCQ (Single Correct Answer) Marks: +1 / -0.33

Let $U = \{1, 2, \dots, n\}$. Let $A = \{(x, X) \mid x \in X, X \subseteq U\}$. Consider the following two statements on $|A|$.

I. $|A| = n2^{n-1}$

II. $|A| = \sum_{k=1}^n k \binom{n}{k}$

Which of the above statements is/are TRUE?

- (A) Neither I nor II
 (B) Only II
 (C) Both I and II
 (D) Only I

9. GATE CSE 2019

MCQ (Single Correct Answer) Marks: +1 / -0.33

Let G be an arbitrary group. Consider the following relations on G :

$$R_1: \forall a, b \in G, aR_1b \text{ if and only if } \exists g \in G \text{ such that } a = g^{-1}bg$$

$$R_2: \forall a, b \in G, aR_2b \text{ if and only if } a = b^{-1}$$

Which of the above is/are equivalence relation/relations?

- (A) R_1 only
 (B) Neither R_1 nor R_2
 (C) R_1 and R_2
 (D) R_2 only

10. GATE CSE 2015 Set 1

MCQ (Single Correct Answer) Marks: +1 / -0.3

For a set A , the power set of A is denoted by 2^A . If $A = \{5, \{6\}, \{7\}\}$, which of the following options are TRUE?

I. $\phi \in 2^A$

II. $\phi \subseteq 2^A$

III. $\{5, \{6\}\} \in 2^A$

IV. $\{5, \{6\}\} \subseteq 2^A$

- (A) I and III only
 (B) II and III only
 (C) I, II and III only
 (D) I, II and IV only

11. GATE CSE 2015 Set 3

MCQ (Single Correct Answer) Marks: +1 / -0.3

Suppose U is the power set of the set $S = \{1, 2, 3, 4, 5, 6\}$. For any $T \in U$, let $|T|$ denote the number of elements in T and T' denote the complement of T . For any $T, R \in U$, let $T \setminus R$ be the set of all elements in T which are not in R . Which one of the following is true?

- (A) $\forall X \in U (|X| = |X'|)$
 (B) $\exists X \in U \exists Y \in U (|X| = 5, |Y| = 5 \text{ and } X \cap Y = \phi)$
 (C) $\forall X \in U \forall Y \in U (|X| = 2, |Y| = 3 \text{ and } X \setminus Y = \phi)$
 (D) $\forall X \in U \forall Y \in U (X \setminus Y = Y' \setminus X')$

12. GATE CSE 2015 Set 2

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let R be the relation on the set of positive integers such that aRb if and only if a and b are distinct and have a common divisor other than 1. Which one of the following statements about R is true?

- (A) R is symmetric and reflexive but not transitive
 (B) R is reflexive but not symmetric and not transitive
 (C) R is transitive but not reflexive and not symmetric
 (D) R is symmetric but not reflexive and not transitive

13. GATE CSE 2014 Set 3

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let X and Y be finite sets and $f: X \rightarrow Y$ be a function. Which one of the following statements is TRUE?

- (A) For any subsets A and B of X , $|f(A \cup B)| = |f(A)| + |f(B)|$
 (B) For any subsets A and B of X , $f(A \cap B) = f(A) \cap f(B)$
 (C) For any subsets $|f(A \cap B)| = \min\{|f(A)|, |f(B)|\}$
 (D) for any subsets S and T of Y , $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$

14. GATE CSE 2013

MCQ (Single Correct Answer) Marks: +1 / -0.3

A Binary operation $\oplus$ on a set of integers is defined as $x \oplus y = x^2 + y^2$. Which one of the following statements is **TRUE** about $\oplus$?

- (A) Commutative but not associative
 (B) Both commutative and associative
 (C) Associative but not Commutative
 (D) Neither commutative nor associative

15. GATE CSE 2013

MCQ (Single Correct Answer) Marks: +1 / -0.3

Which one of the following functions is continuous at $x = 3$?

- (A) $f(x) = \begin{cases} 2, & \text{if } x = 3 \\ x - 1 & \text{if } x > 3 \\ \frac{x+3}{3}, & \text{if } x < 3 \end{cases}$
- (B) $f(x) = \begin{cases} 4, & \text{if } x = 3 \\ 8 - x & \text{if } x \neq 3 \end{cases}$
- (C) $f(x) = \begin{cases} x + 3, & \text{if } x \leq 3 \\ x - 4 & \text{if } x > 3 \end{cases}$
- (D) $f(x) = \frac{1}{x^3 - 27}, \text{ if } x \neq 3$

16. GATE CSE 2010

MCQ (Single Correct Answer) Marks: +1 / -0.3

Consider the set $S = \{1, \omega, \omega^2\}$, where ω and ω^2 , are cube roots of unity. If $*$ denotes the multiplication operation, the structure $\{S, *\}$ forms

- (A) a group
- (B) a ring
- (C) an integral domain
- (D) a field

17. GATE CSE 2010

MCQ (Single Correct Answer) Marks: +1 / -0.3

What is the possible number of reflexive relations on a set 5 elements?

- (A) 2^{10}
- (B) 2^{15}
- (C) 2^{20}
- (D) 2^{25}

18. GATE CSE 2009

MCQ (Single Correct Answer) Marks: +1 / -0.3

consider the binary relation $R = \{(x, y), (x, z), (z, x), (z, y)\}$ on the set $\{x, y, z\}$. which one of the following is TRUE?

- (A) R is symmetric but *NOT* antisymmetric
- (B) R is NOT symmetric but antisymmetric
- (C) R is both symmetric and antisymmetric.
- (D) R is neither symmetric nor antisymmetric.

19. GATE CSE 2008

MCQ (Single Correct Answer) Marks: +1 / -0.3

If P, Q, R are subsets of the universal set U , then $(P \cap Q \cap R) \cup (P^c \cap Q \cap R) \cup Q^c \cup R^c$ is

- (A) $Q^c \cup R^c$
- (B) $P \cup Q^c \cup R^c$
- (C) $P^c \cup Q^c \cup R^c$
- (D) U

20. GATE CSE 2007

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let S be a set of n elements. The number of ordered pairs in the largest and the smallest equivalence relations on S are

- (A) n and n
- (B) n^2 and n
- (C) n^2 and 0
- (D) n and 1

21. GATE CSE 2006

MCQ (Single Correct Answer) Marks: +1 / -0.3

The set $\{1, 2, 3, 5, 7, 8, 9\}$ under multiplication modulo 10 is not a group. Given below are four plausible reasons.

Which one of them is false?

- (A) It is not closed
- (B) 2 does not have an inverse
- (C) 3 does not have an inverse
- (D) 8 does not have an inverse

22. GATE CSE 2006

MCQ (Single Correct Answer) Marks: +1 / -0.3

A relation R is defined on ordered pairs of integers as follows: $(x, y)R(u, v)$ if $x < u$ and $y > v$. Then R is

- (A) Neither a Partial Order nor an Equivalence Relation
- (B) A Partial Order but not a Total Order
- (C) A Total Order
- (D) An Equivalence relation

23. GATE CSE 2006

MCQ (Single Correct Answer) Marks: +1 / -0.3

For the set N of natural numbers and a binary operation $f : N \times N \rightarrow N$, an element $z \in N$ is called an identity for f if $f(a, z) = a = f(z, a)$ for all $a \in N$. Which of the following binary operations have an identity?

- I) $f(x, y) = x + y - 3$
- II) $f(x, y) = \max(x, y)$
- III) $f(x, y) = x^y$

- (A) I and II only
- (B) II and III only
- (C) I and III only
- (D) None of them

24. GATE CSE 2006

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let X, Y, Z be sets of sizes x, y and z respectively. Let $W = X \times Y$ and E be the set of all subjects of W . The number of functions from Z to E is

- (A) $2^{2^{xy}}$
 (B) 2×2^{xy}
 (C) $2^{2^{x+y}}$
 (D) 2^{xyz}

25. GATE CSE 2005

MCQ (Single Correct Answer) Marks: +1 / -0.3

The set $\{1, 2, 4, 7, 8, 11, 13, 14\}$ is a group under multiplication modulo 15. The inverse of 4 and 7 are respectively:

- (A) 3 and 13
 (B) 2 and 11
 (C) 4 and 13
 (D) 8 and 14

26. GATE CSE 2005

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let f be a function from a set A to a set B , g a function from B to C , and h a function from A to C , such that $h(a) = g(f(a))$ for all $a \in A$. Which of the following statements is always true for all such functions f and g ?

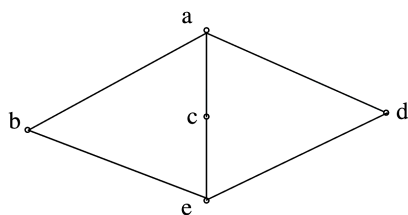
- (A) g is onto $\Rightarrow h$ is onto
 (B) h is onto $\Rightarrow f$ is onto
 (C) h is onto $\Rightarrow g$ is onto
 (D) h is onto $\Rightarrow f$ and g are onto

27. GATE CSE 2005

MCQ (Single Correct Answer) Marks: +1 / -0.3

The following is the Hasse diagram of the poset $[\{a, b, c, d, e\}, \prec]$

The poset is:



- (A) not a lattice
 (B) a lattice but not a distributive lattice
 (C) a distributive lattice but not a Boolean algebra
 (D) a Boolean algebra

28. GATE CSE 2005

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let A, B and C be non-empty sets and let $X = (A - B) - C$ and $Y = (A - C) - (B - C)$

Which one of the following is TRUE?

- (A) $X = Y$
 (B) $X \subset Y$
 (C) $Y \subset X$
 (D) None of these

29. GATE CSE 2004

MCQ (Single Correct Answer) Marks: +1 / -0.3

Consider the binary relation: $S = \{(x, y) \mid y = x + 1 \text{ and } x, y \in \{0, 1, 2, \dots\}\}$

The reflexive transitive closure of S is

- (A) $\{(x, y) \mid y > x \text{ and } x, y \in \{0, 1, 2, \dots\}\}$
 (B) $\{(x, y) \mid y \geq x \text{ and } x, y \in \{0, 1, 2, \dots\}\}$
 (C) $\{(x, y) \mid y < x \text{ and } x, y \in \{0, 1, 2, \dots\}\}$
 (D) $\{(x, y) \mid y \leq x \text{ and } x, y \in \{0, 1, 2, \dots\}\}$

30. GATE CSE 2004

MCQ (Single Correct Answer) Marks: +1 / -0.3

The number of different $n \times n$ symmetric matrices with each element being either 0 or 1 is (Note: power $(2, x)$ is same as 2^x)

- (A) power $(2, n)$
 (B) power $(2, n^2)$
 (C) $(2, (n^2 + n)/2)$
 (D) power $(2, (n^2 - n)/2)$

31. GATE CSE 2004

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let R_1 be a relation from $A = \{1, 3, 5, 7\}$ to $B = \{2, 4, 6, 8\}$ and R_2 be another relation from B to $C = \{1, 2, 3, 4\}$ as defined below:

- i) An element x in A is related to an element y in B (under R_1) if $x + y$ is divisible by 3.
 ii) An element EE in B is related to an elements y in C (under R_2) if $x + y$ is even but not divisible by 3.

Which is the composite relation $R_1 R_2$ from A to C ?

- (A) $R_1 R_2 = \{(1, 2), (1, 4), (3, 3), (5, 4), (7, 3)\}$
 (B) $R_1 R_2 = \{(1, 2), (1, 3), (3, 2), (5, 2), (7, 3)\}$
 (C) $R_1 R_2 = \{(1, 2), (3, 2), (3, 4), (5, 4), (7, 2)\}$
 (D) $R_1 R_2 = \{(3, 2), (3, 4), (5, 1), (5, 3), (7, 1)\}$

32. GATE CSE 2001

MCQ (Single Correct Answer) Marks: +1 / -0.3

Consider the following relations:

 $R_1(a, b) \text{ iff } (a + b) \text{ is even over the set of integers}$ $R_2(a, b) \text{ iff } (a + b) \text{ is odd over the set of integers}$ $R_3(a, b) \text{ iff } a.b > 0 \text{ over the set of non-zero rational numbers}$ $R_4(a, b) \text{ iff } |a - b| \leq 2 \text{ over the set of natural numbers}$

Which of the following statements is correct?

- (A) R_1 and R_2 are equivalence relations, R_3 and R_4 are not
 (B) R_1 and R_3 are equivalence relations, R_2 and R_4 are not
 (C) R_1 and R_4 are equivalence relations, R_2 R_3 are not
 (D) R_1, R_2, R_3 and R_4 are all equivalence relations

33. GATE CSE 1997

MCQ (Single Correct Answer) Marks: +1 / -0.3

The number of equivalence relations on the set $\{1, 2, 3, 4\}$ is

- (A) 15
 (B) 16
 (C) 24
 (D) 4

34. GATE CSE 1996

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let A and B be sets and let A^c and B^c denote the complements of the sets A and B . The set $(A - B) \cup (B - A) \cup (A \cap B)$ is equal to

- (A) $A \cup B$
 (B) $A^c \cup B^c$
 (C) $A \cap B$
 (D) $A^c \cap B^c$

35. GATE CSE 1996

MCQ (Single Correct Answer) Marks: +1 / -0.3

Which of the following statements is false?

- (A) The set of rational numbers is an abelian group under addition.
 (B) The set of integers is an abelian group under addition.
 (C) The set of rational numbers from an abelian group under multiplication.
 (D) The set of real numbers excluding zero is an abelian group under multiplication.

36. GATE CSE 1996

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let $X = \{2, 3, 6, 12, 24\}$. Let $\leq$ the partial order defined by $x \leq y$ if x divides y . The number of edges in the Hasse diagram of $(X, \leq)$ is

- (A) 3
 (B) 4
 (C) 9
 (D) None of the above

37. GATE CSE 1995

MCQ (Single Correct Answer) Marks: +1 / -0.3

Let R be a symmetric and transitive relation on a set A . Then

- (A) R is reflexive and hence an equivalence relation.
 (B) R is reflexive and hence partial order.
 (C) R is not reflexive and hence not an equivalence relation.
 (D) None of the above.

38. GATE CSE 1987

Subjective Marks: +1 / -0

(a) How many binary relations are there on a set A with n elements?(b) How many one - to - one functions are there from a set A with n elements onto itself